**This assignment will not be graded and is only for practice.**

## Level 1

1) The null space of a matrix $A_{4\times 3}$ is **1 point**

- the subspace $W = \{x \in \mathrm{R}^4 \mid Ax = 0\}$ of $\mathrm{R}^4$.
- the subspace $W = \{x \in \mathrm{R}^3 \mid Ax = 0\}$ of $\mathrm{R}^3$.
- the subspace $W = \{x \in \mathrm{R}^2 \mid Ax = 0\}$ of $\mathrm{R}^2$.
- the first column of the matrix $A_{4\times 3}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

the subspace $W = \{x \in \mathrm{R}^3 \mid Ax = 0\}$ of $\mathrm{R}^3$.

2) Nullity of a matrix $A_{3\times 4}$ is **1 point**

- 3
- 4
- the number of independent variables in the system of linear equations $Ax = 0$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$.
- the number of dependent variables in the system of linear equations $Ax = 0$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

the number of independent variables in the system of linear equations $Ax = 0$, where $x =$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}.$$

Consider the system of linear equations $Ax = 0$, where $x = [x_1 \quad x_2 \quad x_3]^T$ and $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Use the above information to answer questions 3, 4 and 5.

3) Choose the correct statement from the following options. ***1 point***

- ○ $x_1, x_2$ and $x_3$ all are dependent variables.
- ○ $x_2$ is the only dependent variable.
- ○ $x_1$ and $x_3$ are the dependent variables and $x_2$ is the only independent variable.
- ○ $x_1, x_2$ and $x_3$ all are independent variables.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x_1$ and $x_3$ are the dependent variables and $x_2$ is the only independent variable.

4) Which one of the following vector spaces represents the null space of $A$? ***1 point***

- ○ $\{(t_1, 0, t_2) \mid t_1, t_2 \in \mathrm{R}\}$
- ○ $\{(0, t, 0) \mid t \in \mathrm{R}\}$
- ○ $\{(t_1, t_2, 0) \mid t_1, t_2 \in \mathrm{R}\}$
- ○ $\{(0, 0, t) \mid t \in \mathrm{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(0, t, 0) \mid t \in \mathrm{R}\}$

5) What will be the nullity of $A$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

***1 point***

6) Find the nullity of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{bmatrix}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

***1 point***

7) Let $S$ denote the set of solutions of the homogeneous system of linear equations $Ax = 0$. Which of the following statements is (are) true? ***1 point***

- [ ] If $x_1$ and $x_2$ are in $S$, then $x_1 - x_2$ is in $S$.
- [ ] If $x$ is in $S$, then $cx$ is in $S$, for any $c \in \mathrm{R}$.
- [ ] If $x_1$ and $x_2$ are in $S$, then $\alpha x_1 + \beta x_2$ is in $S$, for any $\alpha, \beta \in \mathrm{R}$.
- [ ] If $x$ is in $S$, then $A^n x = 0$ for any $n \in \mathrm{N} \setminus \{0\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $x_1$ and $x_2$ are in $S$, then $x_1 - x_2$ is in $S$.
If $x$ is in $S$, then $cx$ is in $S$, for any $c \in \mathrm{R}$.
If $x_1$ and $x_2$ are in $S$, then $\alpha x_1 + \beta x_2$ is in $S$, for any $\alpha, \beta \in \mathrm{R}$.
If $x$ is in $S$, then $A^n x = 0$ for any $n \in \mathrm{N} \setminus \{0\}$.

8) Find out the value of $a$ for which the matrix $\begin{bmatrix} 1 & 2 \\ 3 & a \end{bmatrix}$ has nullity 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

***1 point***

**Level 2**

9) Consider the coefficient matrix $A$ of the following system of linear equations: ***1 point***

$$x_1 + x_2 + x_4 = 0$$
$$x_2 + x_3 = 0$$
$$x_1 - x_3 + x_4 = 0$$

Which one of the following vector spaces represents the null space of $A$?

- ○ $\{(t_1 + t_2, t_1, t_1, t_2) \mid t_1, t_2 \in \mathrm{R}\}$
- ○ $\{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in \mathrm{R}\}$
- ○ $\{(t_1 + t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in \mathrm{R}\}$
- ○ $\{(t_1 - t_2, -t_1, t_1, -t_2) \mid t_1, t_2 \in \mathrm{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in \mathrm{R}\}$

10) Find the nullity of the matrix $A$, where $A = [a_{ij}]$ is of order $3 \times 3$ and $a_{i,j} = \min\{i, j\}$, $i, j = 1, 2, 3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

***1 point***

11) Choose the correct set of options from the following. ***1 point***

- ☐ The nullity of a non-zero scalar matrix of order 3 must be 3.
- ☐ The nullity of a non-zero scalar matrix of order 3 must be 0.
- ☐ The nullity of a non-zero diagonal matrix of order 3 must be 3.
- ☐ The nullity of a non-zero diagonal matrix of order 3 can be at most 2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The nullity of a non-zero scalar matrix of order 3 must be 0.
The nullity of a non-zero diagonal matrix of order 3 can be at most 2.

12) Consider the system $Ax = b$, where $A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$. The number of solutions of when $b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

***1 point***

**Your score is: 0/12**